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Catalytic conversion of bioethanol to value-added chemicals and fuels: A review

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ABSTRACT

Bioethanol produced *via* valorisation of renewable biomass is of great interest to many industries. The increased availability and decreased cost of bioethanol make it a promising platform molecule to produce a wide range of value-added chemicals and fuels *via* the catalytic conversions. This paper provides a comprehensive review of catalytic conversions of bioethanol to a variety of chemicals/fuels such as hydrogen, C_2–C_4 olefins, gasoline and small oxygenates. Specifically, the focus was placed on the relationship between the catalyst property (such as pore structure, acidity, active metal sites, and catalyst supports) and the catalytic performance (including catalyst activity and stability), as well as the reaction mechanisms involved. Future research avenues on the catalyst design for improving catalytic valorisation of bioethanol are also discussed.

1. Introduction

The rising demand for energy around the world caused the extensive use of fossil fuels, leading to environmental issues, such as climate changes, global warming, and air pollution [1,2]. Therefore, the development of renewable and sustainable energy sources has attracted great attention from the scientific community. Biofuels, such as biodiesel and bioethanol, produced from biomass valorisation, are now considered as promising alternatives to the fossil fuels due to their environmental-friendly and sustainable features. In the United States (US), the Energy Independence and Security Act announced that the renewable fuels used by transportation need to increase to 36 billion gallons per year by 2022 [3,4]. The Biofuels Research advisory Council of European Union (EU) proposed increasing the use of biofuels in transportation to 25% by 2030 [5]. The biofuel policies in US and EU aim to reduce greenhouse gas emissions, improve the life cycle energy efficiency of biofuels, and promote the national biofuel industry (*i.e.,* the production of biofuels including bioethanol, biobutanol, and biodiesel). In Asia, the production and use of biofuels are also widely encouraged. For example, in 2017, the Chinese government announced to roll out the mandatory blending of 10% ethanol in gasoline nationally by 2020 [6]. To meet the target, significant development has been made to enhance the bioethanol production capacity. In Association of Southeast Asian Nation (ASEAN), biofuel policies are robust because domestic biofuel consumption is a means for energy security while promoting socio-economic development through ensuring the demand for strategically critical agricultural commodities [7]. As one of the ASEAN member states, Thailand has also set a target to increase bioethanol production from 6.2 million litres per day in 2016 to 9.0 million litres per day in 2022 to reduce greenhouse gas emission and fossil fuel dependency [8]. The world production of bioethanol was 110 billion litres (BL) in 2018 and is expected to reach 140 BL in 2022 [9]. In addition, the market of bioethanol fuel has shown a dramatic increase and growing at a compound annual growth rate of 7.6% from 2016 to 2022.

Abbreviations: SR, Steam reforming; ESR, Ethanol steam reforming; WGS, Water gas shift; TPD, Temperature programmed desorption; AC, Activated carbon; CNT, Carbon nanotube; CNF, Carbon nanofibre; ETE, Ethanol-to-ethylene; DEE, Diethyl ether; ETP, Ethanol-to-propylene; MPV, Meerwein-Ponndorf-Verley; BD, 1,3-Butadiene; TOS, Time-on-stream; ETG, Ethanol-to-gasoline; MTG, Methanol-to-gasoline; BTX, Benzene toluene xylene; HAP, Hydroxyapatite; CHAP, Carbonate hydroxyapatite.

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